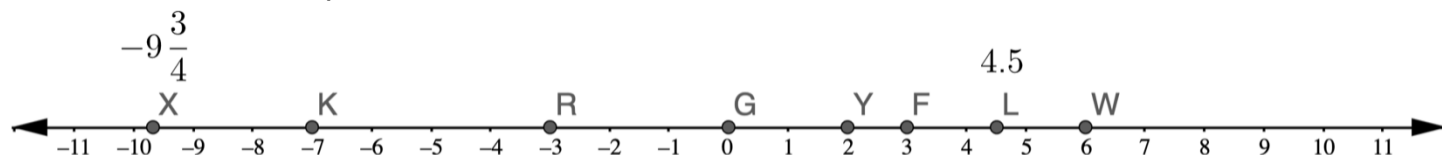


Grade 6 Accelerated
Unit 3
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

A number line with points X , K , R , G , Y , F , L , and W is shown.



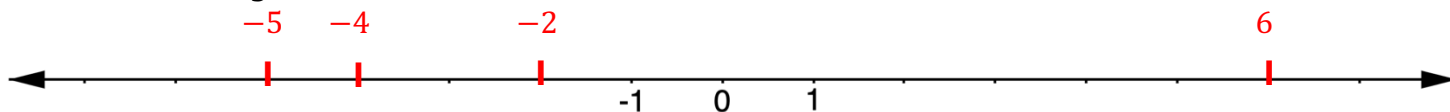
1. Complete the table by identifying which points belong in each category. Some points may be used more than once and some may not be used at all.

Natural Numbers	Whole Numbers	Integers	Rational Numbers
Y F W	G F Y W	K Y R F G W	X Y K F R L G W

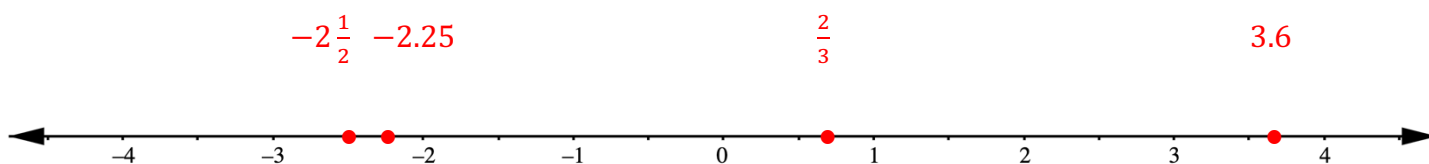
2. Complete the table by identifying which points belong in each category. Some points may be used more than once and some may not be used at all.

Positive Numbers	Negative Numbers
Y F L W	R K X

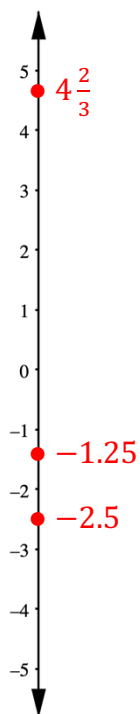
3. Plot the integers 6 , -4 , -2 , and -5 on the number line.



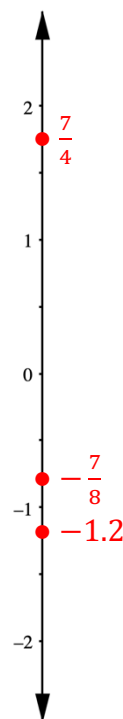
4. Plot the numbers $\frac{2}{3}$, -2.25 , 3.6 , and $-2\frac{1}{2}$ on the number line.



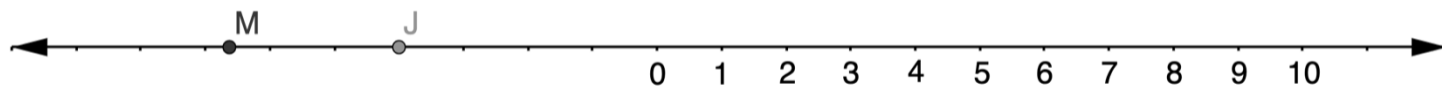
5. Plot the numbers $-\frac{5}{2}$, $4\frac{2}{3}$, and -1.25



6. Plot the numbers $\frac{7}{4}$, $-\frac{7}{8}$, and -1.2



A number line with points M and J is shown.



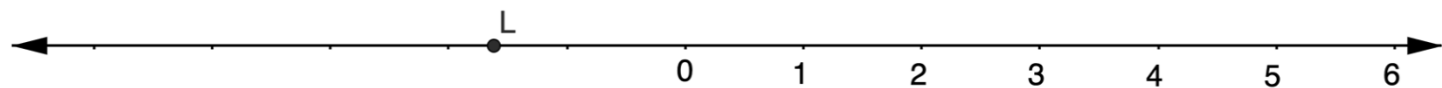
7. Explain how to estimate the value of J .

Count markings, it is roughly 4 units left of zero. It is estimated to be -4 .

8. Explain how to estimate the value of M .

It is between -6 and -7 , slightly passed the midpoint of the two, so it is estimated to be $-6\frac{2}{3}$.

Jimmy and Luna both estimated the value of L on the number line shown.



9. Jimmy says the value of L is about $-2\frac{1}{3}$. Explain whether you agree with Jimmy.

Disagree. It is between -1 and -2 , so it cannot be left of -2 .

10. Luna says the value of L is about $1\frac{2}{3}$. Explain whether you agree with Luna.

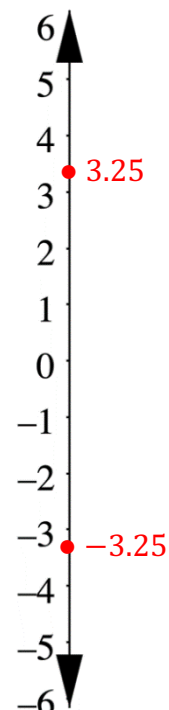
Disagree. The point is to the left of zero so it must be negative.

Use the vertical number line to answer the following problems.

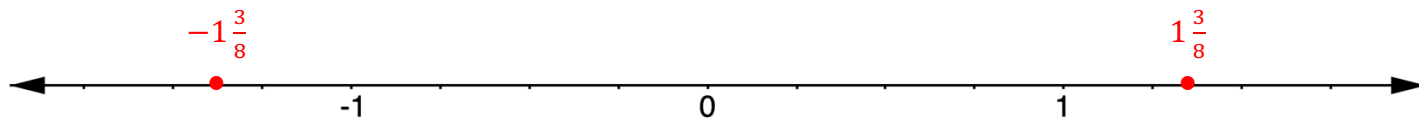
1. Plot 3.25 and its opposite.

2. Which is larger, 3.25 or its opposite?

3.25 because it is positive.
 $3.25 > -3.25$



Use the horizontal number line to answer the following problems.



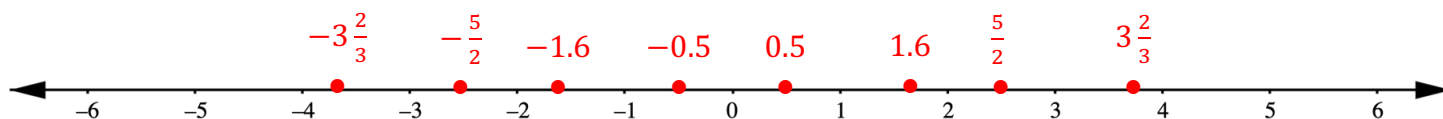
3. Plot $-1\frac{3}{8}$ and its opposite.

4. Which is larger, $-1\frac{3}{8}$ or its opposite?

Its opposite as it is positive.

$$1\frac{3}{8} > -1\frac{3}{8}$$

5. Plot and label each value and its opposite on the number line: -1.6 , $3\frac{2}{3}$, 0.5 , $-\frac{5}{2}$



Complete each comparison statement for the pairs of numbers you plotted in question 5. Use $<$ or $>$ in the first blank and the given number's opposite in the second blank.

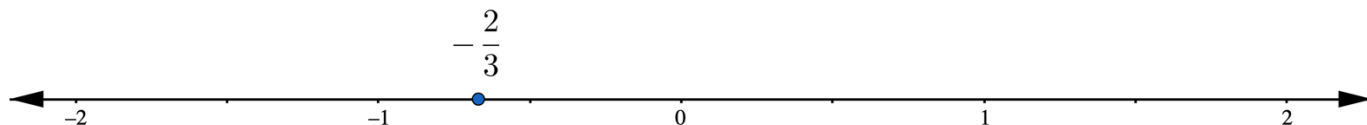
6. $-1.6 < 1.6$

7. $3\frac{2}{3} > -3\frac{2}{3}$

8. $0.5 > -0.5$

9. $-\frac{5}{2} < \frac{5}{2}$

10. A number line with the value $-\frac{2}{3}$ plotted is shown.



Stefan said that the opposite of $-\frac{2}{3}$ is $\frac{1}{3}$ because it's on the other side of 0. Explain whether you agree with Stefan.

Disagree, although it is on the other side of zero its value is not the exact same distance away from zero on the number line so they are not opposites.

Explain why each inequality statement is true or false.

1. $-0.45 > -1.2$

True, -0.45 is greater than -1.2 , it is further right on a number line.

2. $72\% < -8\%$

False, 72% is greater than -8% , -8% is farther left on a number line and is therefore less.

Complete each statement with an inequality symbol.

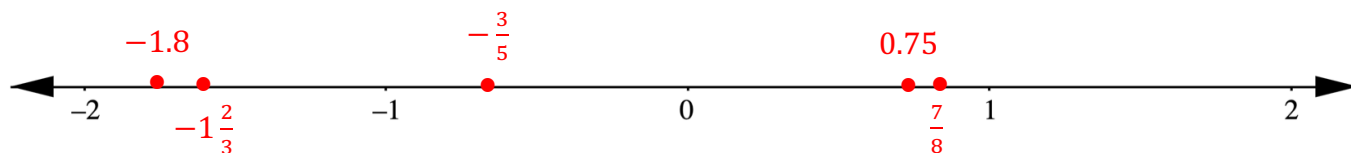
3. $12.3 > 125\%$

4. $-9.45 > -9.88$

A set of rational numbers is shown.

$$-1\frac{2}{3}, \quad -1.8, \quad \frac{7}{8}, \quad -\frac{3}{5}, \quad 0.75$$

5. Plot each of the values on the number line. Label each point with its numeric value.



6. Complete the inequality statement to show the relationship between the values.

$$-1.8 < -1\frac{2}{3} < -\frac{3}{5} < 0.75 < \frac{7}{8}$$

Complete each statement using $<$, $>$, or $=$.

7. $-\frac{7}{5} = -1\frac{2}{5}$

8. $-1.4 > -\frac{3}{2}$

9. Order the values from least to greatest.

$$-2.\bar{6}, \frac{2.75}{4}, -\frac{4.5}{2}, -\frac{0.6}{5}, 0, 2.\bar{3}$$

$$-\frac{9}{2}, -2\frac{2}{3}, -\frac{3}{5}, 0, 2\frac{1}{3}, \frac{11}{4}$$

10. Order the values from least to greatest.

$$1\frac{2}{3}, -3.8, 2.6$$

$$-3.75, \frac{5}{3}, -3\frac{4}{5}, -2.6, 2\frac{3}{5}$$

$$-3\frac{4}{5}, -3.75, -2.6, \frac{5}{3}, 2\frac{3}{5}$$

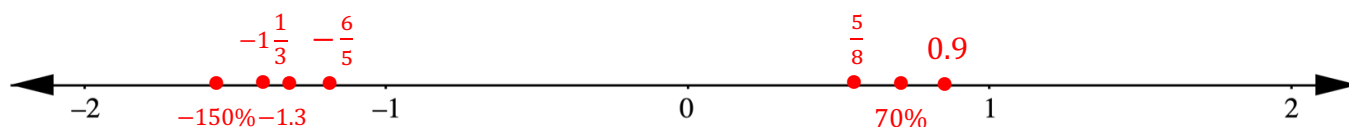
Grade 6 Accelerated
Unit 3
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

A set of rational numbers is shown.

$$-\frac{6}{5}, \quad -1.3, \quad 70\%, \quad -1\frac{1}{3}, \quad \frac{5}{8}, \quad -150\%, \quad 0.9$$

1. Plot each of the values on the number line. Label each point with its numeric value.



2. Order the values from least to greatest.

$$-150\%, -1\frac{1}{3}, -1.3, -\frac{6}{5}, \frac{5}{8}, 70\%, 0.9$$

Complete each statement using $<$, $>$, or $=$.

3. $-8.88 > -8\frac{9}{10}$

4. $2.4\% < \frac{1}{5}$

5. $-67\% < 0.67$

6. $-52\% = -\frac{13}{25}$

7. $-3.472 < -3\frac{5}{12}$

8. $-12\% > -1\frac{1}{5}$

9. Order the values from least to greatest.

$$-\frac{5}{9}, \quad 65\%, \quad -\frac{3}{10}, \quad -0.451, \quad 0.815, \quad \frac{3}{5}$$

$$-\frac{5}{9}, -0.451, -\frac{3}{10}, \frac{3}{5}, 65\%, 0.815$$

10. Luka wrote the inequality statement:

$$-4.111 < -450\% < -\frac{14}{3}$$

Explain whether you agree with Luka.

Luka is incorrect, his order is backwards because $-\frac{14}{3}$ is furthest from 0, then -450% , and closest to 0 is -4.111 .

$$-\frac{14}{3} < -450\% < -4.111$$

Grade 6 Accelerated
Unit 3
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Write a value that represents each description.

1. A mountain climber standing 12,840 feet above sea level

12,840 or +12,840

2. A temperature decrease of 3.5°

-3.5°

3. A deposit of \$372.65 to a savings account

372.65 or +372.65

4. An expense of \$49.33

-49.33

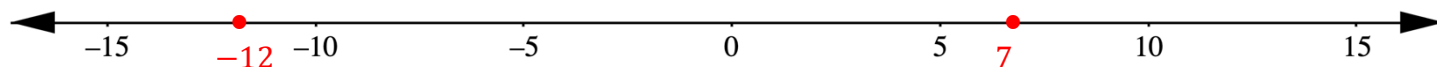
In the last two plays of a football game, the offense had a gain of 7 yards and then a loss of 12 yards.

5. Write a rational number that represents the football's change in position with each play.

Gain of 7 yards: 7

Loss of 12 yards: -12

6. Plot both rational numbers on the number line.



7. What does 0 represent in this context?

No gains or losses in yards

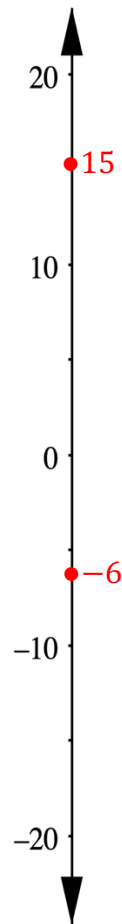
A boat is on the water in Crystal Keys. The top of the mast of the boat is 15 feet above sea level, and the bottom of the boat is 6 feet below sea level.

8. Write a rational number that represents the elevation of each part of the boat.

Top of the mast: 15

Bottom of the boat: -6

9. Plot both rational numbers on the number line.



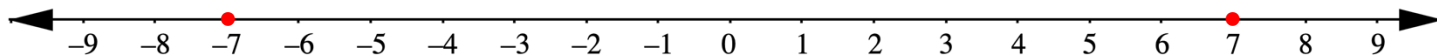
10. What does 0 represent in this context?

Sea level

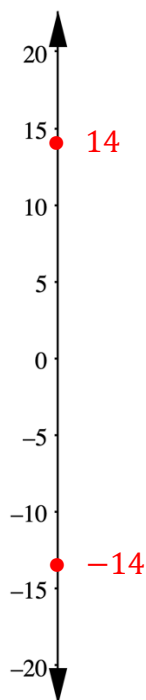
Grade 6 Accelerated
Unit 3
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Plot 7 and its opposite on the number line.



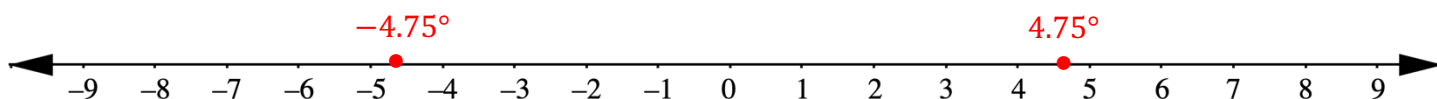
2. On the number line, plot a location of 14 feet above sea level and its opposite.



What is the meaning of 0 in this context?

Sea level

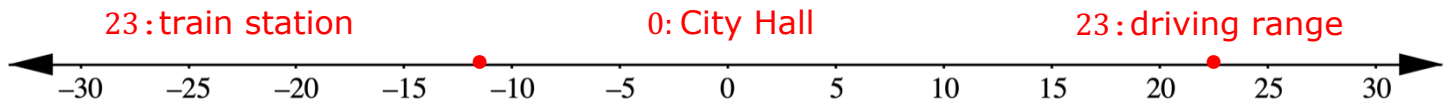
3. Plot a decrease of 4.75° and its opposite on the number line. What is the meaning of 0 in this context?



No change in temperature at 0.

From City Hall, the driving range is 23 blocks East and the train station is 23 blocks West.

4. Model all three locations on the number line.



5. What does 0 represent in this situation?

City Hall

6. From City Hall, is it farther to the driving range or to the train station?

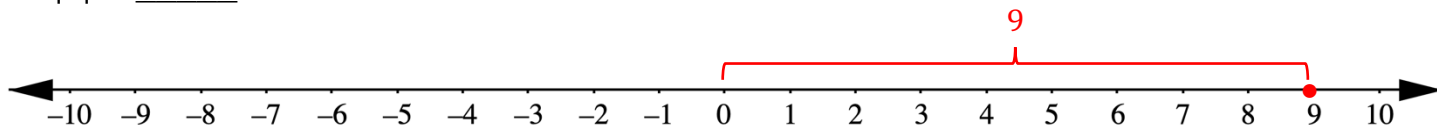
They are equal distances from City Hall.

Complete the table for each description.

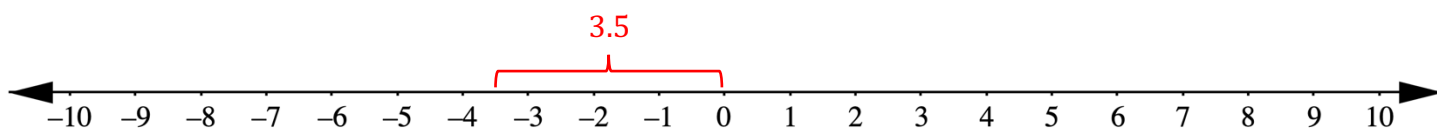
	Description	Rational Number	Opposite	Meaning of Opposite	Meaning of 0
7.	A withdrawal of \$94.72	-94.72	94.72	A deposit of 94.72	No transactions
8.	A gain of 1.2 pounds	1.2	-1.2	A loss of 1.2 pounds	No weight change
9.	397 yards above sea level	397	-397	397 yards below sea level	At sea level
10.	A temperature decrease of 9.2°	-9.2	9.2	A temperature gain of 9.2°	No temperature change

Complete each statement and model it on the number line.

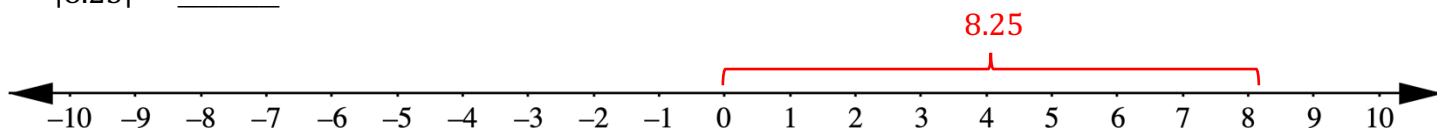
1. $|9| = \underline{9}$



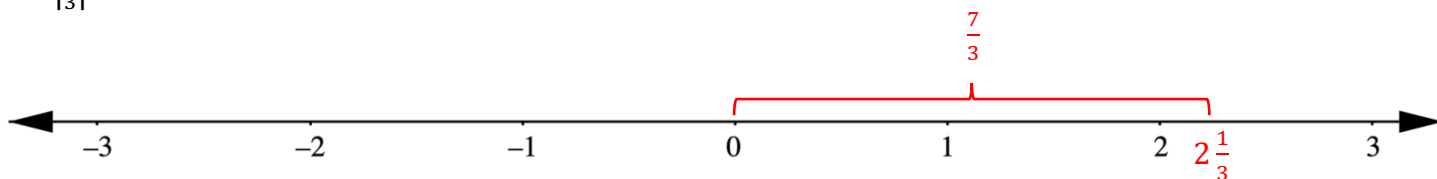
2. $|-3.5| = \underline{3.5}$



3. $|8.25| = \underline{8.25}$



4. $\left|\frac{7}{3}\right| = \underline{\frac{7}{3}}$



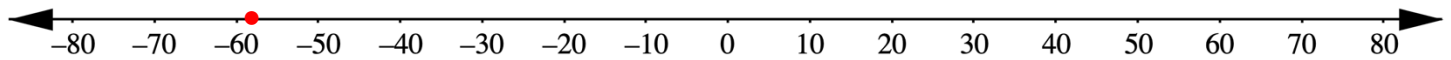
5. Martina claims that an absolute value cannot be negative. Explain whether you agree with Martina.

This is correct, distance is measured in positive units and absolute value is distance from 0.

Raechelle makes a withdrawal of \$59.74 from her checking account.

6. Model Raechelle's transaction on the number line.

-59.74



7. What does the absolute value of Raechelle's transaction mean in this context?

The absolute value is the amount of money involved in the transaction.

8. From noon to midnight, the temperature change in Lakeland, FL was -8.5°F . What does the value $|-8.5|^{\circ}\text{F}$ mean in this context?

That the degrees of temperature change is 8.5.

9. Describe the meaning of each value in the context of elevation.

Value	Interpretation
5,362 feet	5,362 ft above sea level
-743 feet	743 ft below sea level
$ 67 $ feet	67 ft above sea level
$ -3,918 $ feet	3,918 ft below sea level

10. The absolute value of Amarya's account balance is \$75.88. What could be Amarya's account balance?

It could be \$75.88 or $-\$75.88$

Grade 6 Accelerated
Unit 3
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Use the symbols $<$, $>$, and $=$ to compare each pair of values.

1. $|-83|$ $>$ -77 $83 > -77$

2. 31.8 $=$ $|-31.8|$ $31.8 = 31.8$

Find the value of each expression.

3. $|-44| - |-18|$
 $44 - 18 = 26$
 34^{14}
 $\frac{-18}{26}$

4. $|-5| \times |-9|$
 $5 \times 9 = 45$

5. $|-23| + |-7| + 2$
 $23 + 7 + 2$
 $30 + 2$
 $= 32$

Sia made two transactions on her bank account last week. She made a withdrawal of \$184.60 and a deposit of \$98.55.

6. Write a value that could be used to describe each of Sia's transactions.

Transaction	Value
Withdrawal	-184.60
Deposit	98.55

7. Use an inequality symbol, $>$ or $<$, to compare the absolute values of Sia's transactions.

$$-|184.60| \text{ } > \text{ } |98.55|$$

8. What does this inequality statement indicate about the total change in Sia's bank account last week?

It indicates that she withdrew more than she deposited.

On April 6th, 2020, two baby giraffes were born at the Miami zoo to two different female giraffes, Mia and Zuri. Mia's calf was 49 pounds above the average newborn giraffe weight, while Zuri's calf was 13 pounds below the average weight.

9. Write a value that could be used to describe each calf's weight in relation to the average.

Calf	Value
Mia's calf	49
Zuri's calf	-13

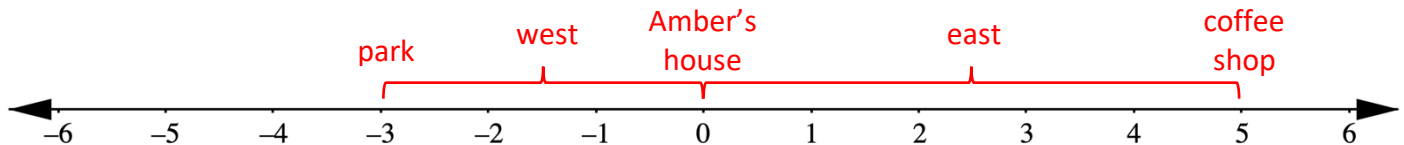
10. Use an inequality symbol, $>$ or $<$, to compare the absolute values.

$$|49| \text{ } > \text{ } |-13|$$



The coffee shop is 5 blocks East of Amber's house. The park is 3 blocks West of Amber's house.

1. Model Amber's house, the coffee shop, and the park on the number line.



2. Write a mathematical statement using absolute value that could be used to find the distance from the coffee shop to the park.

$$|-3| + |5|$$

3. How many blocks is it from the coffee shop to the park?

$$|-3| + |5|$$
$$3 + 5 = 8 \text{ blocks}$$

4. Write a mathematical statement using absolute value that could be used to determine how much farther from Amber's house the coffee shop is than the park.

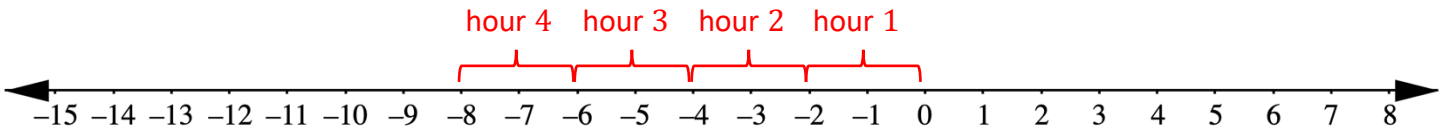
$$|5| - |-3|$$

5. How much farther from Amber's house is the coffee shop than the park?

$$|5| - |-3|$$
$$5 - 3 = 2 \text{ blocks}$$

For the 4 hours between 5pm and 9pm, the temperature change was -2 per hour.

6. Model the change in temperature on the number line.



7. Write a mathematical statement using absolute value that could be used to determine the total decrease in temperature from 5pm to 9pm.

$4 \times |-2|$

In 2020, the country with the greatest increase in population by percent was Niger, with an increase of 3.84%. The country with the greatest decrease was Lithuania, with a decrease of 1.35%.

8. Write a value that could be used to describe each country’s net change in population by percent.

Country	Net Change in Population by Percent
Niger	3.84%
Lithuania	-1.35%

9. Use an inequality symbol, $>$ or $<$, to compare the absolute values of the population changes by percent for each country.

$|3.84\%| > |-1.35\%|$

10. Which country’s population changed more by percentage?
Niger changed more by percentage.